

Cosmos Bio

C-CAMP, Bangalore – 560065



Vision

To pioneer next-generation, evolution-inspired UV bioactives that redefine sun protection for the future of the cosmetic industry.

The UV Filter Market is Structurally Broken – 20 Year Innovation Gap

The industry is trapped in a False Choice between safety risks and poor performance, leaving consumers and regulators without viable alternatives.

Chemical Sunscreens (The Safety Crisis)



Biological Toxicity: Endocrine disruption and confirmed presence in blood & breast milk.



Ecological Damage: Severe Coral Reef Toxicity (Bleaching).



Regulatory Pressure: Subject to increasing global bans.

Examples: Oxybenzone, Avobenzone.

Mineral Sunscreens (The Aesthetic Crisis)



Unusable Aesthetics: Undesirable white, pasty appearance (White Cast).



Poor Sensory Feel: Heavy, greasy feel on the skin.



Safety Concerns: Nanoparticle concerns regarding environmental persistence.

Examples: Titanium Dioxide, Zinc Oxide

WHY THE MARKET IS READY FOR DISRUPTION

Innovation Stagnation

Zero meaningful molecular innovation in over two decades; dependency on legacy filters.

The Clean Beauty Gap

Consumers demand biotech-aligned, safe ingredients but face a lack of alternatives.

Geographic Risk

Supply chains are concentrated in Western players, leaving emerging markets underserved.

Consumers want better. Regulators want safer. Formulators are frustrated. The market is primed for a breakthrough.

Nature Already solved UV Protection. **We Scaled It.**

We are building the first scalable platform for evolution-derived UV bioactives.

Discovery and industrial-scale production of next-generation UV molecules for safe, sustainable photoprotection.

Sustainable UV Protection — Reimagined



Natural biological origin



Absorb and dissipate UV radiation



Photostable



Biocompatible & regulatory-aligned



Multi-functional (anti-inflammatory, hydration)



Effective at low concentrations



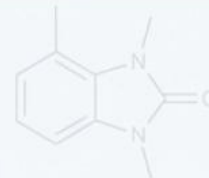
Scalable Fermentation Production

Evolution-Derived UV Bioactives



Ectoïne

Cellular stress protector



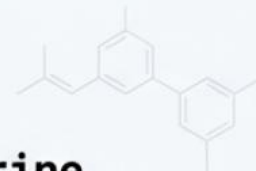
Gadusol

Marine-derived UV absorber



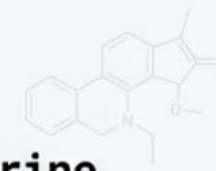
Shinorine

Broad-spectrum MAA



Gadusporine

Next-gen UV molecule



These molecules evolved over millions of years in high-UV environments. We are the first to industrialize them.

Journey from Scientific Vision to Commercial Scale

5 Years of Disciplined Evolution (2020–2025)

PHASE 1: PROOF OF CONCEPT (2020–2021)

Proving it works

2020

Foundation & Vision

Cosmos Biotech LLP incorporated.
Vision set for bioactive UV molecules
& early R&D groundwork.

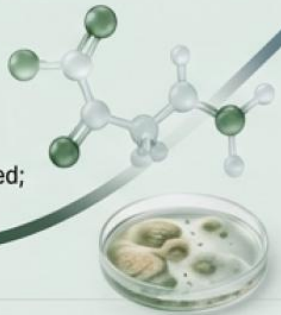
Concept Stage



2021

First Scientific Breakthrough

Ectoine POC established.
Lab-scale biosynthesis demonstrated;
platform feasibility validated.



Scientific Validation

PHASE 2: INSTITUTIONAL VALIDATION (2022–2023)

Securing the backing

2022

BIRAC Grant Awarded

Biotechnology Ignition Grant.
Government-backed funding &
scale-up R&D acceleration.



2023

Structural & Capital Expansion

Cosmos Bio Private Limited incorporated.
Karnataka Elevate Grant &
CCAMP Seed Funding secured.



Non-dilutive Capital & Credibility

PHASE 3: PLATFORM EXPANSION & TRACTION (2024–2025)

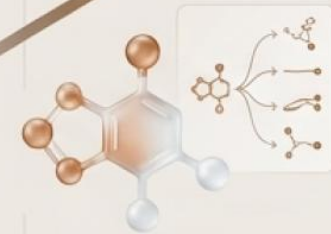
Going to market

2024

Platform Expansion

Gadusol program initiated
(Evolution-derived UV
molecule development).

Multi-molecule strategy



2025

Commercial & Platform Acceleration

- Friends & Family Investment Round (Transition to commercialization).
- Gadusol POC established & Lab-scale biosynthesis validated.
- Pipeline Expansion: Shinorine & Gadusporine programs initiated.
- Commercial Milestone: Samples supplied to:



L'ORÉAL

ESTÉE
LAUDER

CHOSEN
Stores

Market Traction & Brand-Level Engagement

Validated by science. Funded by institutions. Engaged by global brands.

The UV Industry is at an Inflection Point



Regulatory Pressure Is Accelerating

- Increasing scrutiny across US, EU, and APAC
- Reef-safe legislation is expanding globally
- Safety and endocrine concerns are currently under review
- Legacy filters are facing mounting restrictions



Consumer Expectations Have Changed

- Clean beauty is no longer niche—it is mainstream
- A distinct market shift toward biotech and nature-derived ingredients
- Users actively reject greasy textures and white cast
- Demand is high for safe, effective, and elegant formulations



Industry Supply Is Stagnant

- No meaningful new UV molecules in ~20 years
- The synthetic innovation pipeline is dry
- Existing filters significantly limit formulation creativity
- Brands are actively seeking reformulation partners

Strategic Timing for India

Strong national push toward domestic IP in life sciences

A critical opportunity to reduce dependency on Western UV chemistry

First-mover advantage in evolution-derived UV bioactives

The industry cannot stay where it is. The next molecule must be different.

Built to Lead. Designed to Defend.

Foundational IP secured for Ectoine production with a clear roadmap for global protection.



Authority: Controller General of Patents, Designs & Trade Marks

FILED

Application No: 202441050897

Date: July 03, 2024

Title: PRODUCING ECTOINE USING GENETIC ENGINEERING

Foundational filing securing the proprietary biosynthetic optimization.

Full Value-Chain Coverage

Biosynthetic Pathway
Optimization & Synthesis

Production
Scalable Fermentation Methods

Processing
Downstream Purification

Application
Formulation Compatibility

Global Expansion Strategy

Capital allocated to execute PCT filing and multi-jurisdiction strategy (US, EU, Japan, Korea) to ensure comprehensive Freedom-to-Operate (FTO).

Process Moat

Proprietary biosynthetic optimization and scalable fermentation know-how.

Scale Moat

Industrial production capability — not just lab discovery.

Regulatory Moat

Positioning aligned with global safety and clean-beauty trends.

Platform Advantage

Portfolio of evolution-derived molecules — not a single-asset risk.

We are not just inventing molecules — we are industrializing them with protection.

Validation Status

From Discovery to De-Risked Platform

Major Technical Risks Removed

Scientific Feasibility Proven

- ✓ **Biosynthesis**
Lab-scale biosynthesis successfully demonstrated
- ✓ **Purity**
Target compound purity validated against specifications
- ✓ **Efficacy Profile**
Preliminary UV absorption profile confirmed
- ✓ **Resilience**
Stability tested under simulated UV exposure

Core scientific feasibility proven

Validation Path

De-Risking in Progress

Optimizing for Industrial Scale

- ⚙️ **Yield**
Scale-up fermentation yield optimization underway
- ⌚ **Regulatory**
Regulatory dossier preparation in progress
- ⌚ **Photostability**
Long-term photostability studies initiated
- ⌚ **Integration**
Formulation compatibility validation ongoing

*Transitioning from lab validation
to commercial readiness*

“We have crossed scientific feasibility. We are now optimizing for industrial scale and regulatory readiness.”

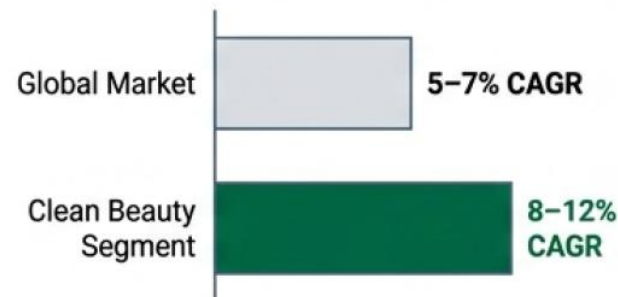
Reinventing the UV Ingredient Layer of a \$15B+ Market

A 'Why Now' opportunity driven by scarcity and regulatory reformulation.

MARKET CONTEXT

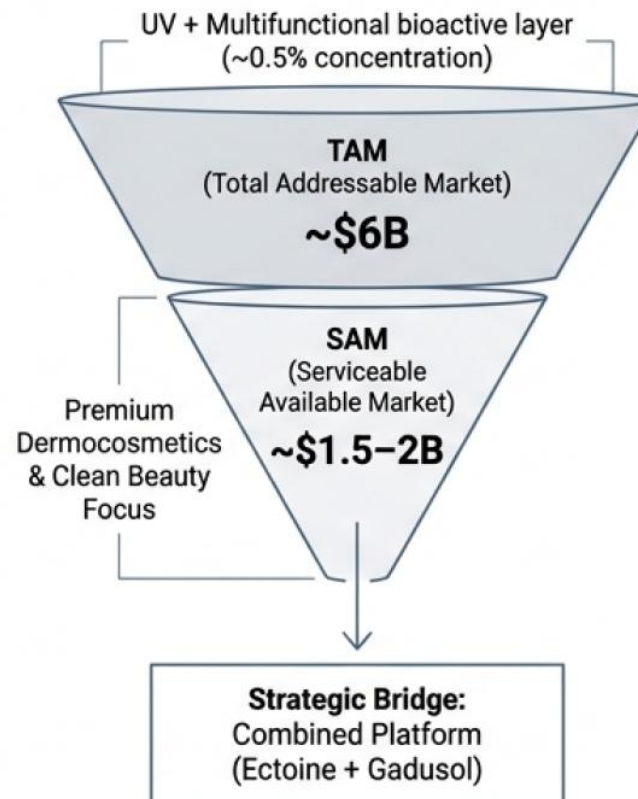
\$15–18B+

Global Sun Care Market



The Driver: Reformulation cycle accelerating due to regulatory pressure; ingredient replacement is currently underway.

MARKET SIZING LOGIC



VALUE REALIZATION

The Moat Section

- ✓ **Scarcity:** No dominant industrial-scale supplier.
- ✓ **Superiority:** Natural + Photostable differentiation.
- ✓ **Positioning:** First-mover advantage.

The Financial Prize Section



Scarcity of scalable Gadusol supply + regulatory-driven reformulation creates a rare window for category leadership.

Landscape

DEFINING THE CATEGORY: Building the First Evolution-Derived UV Bioactive Platform

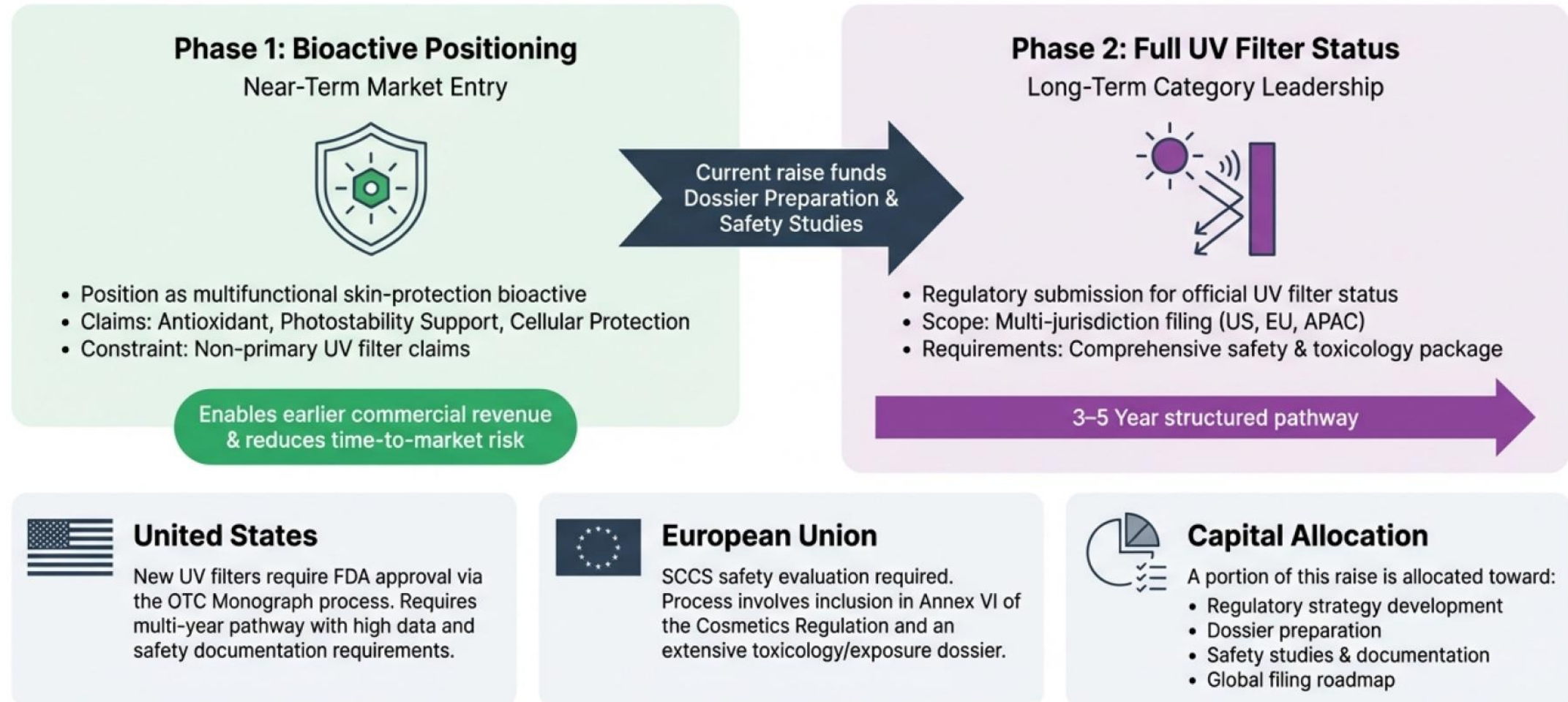
Beyond single molecules—combining scalable biotech with cost leadership

 Bitop AG The Established Monoproducer	 Arcaea The Early-Stage Biotech	 Cosmos The Scalable Platform
Current Strengths - Established global ectoine supplier - Strong brand recognition in dermocosmetics - Proven industrial fermentation	Current Strengths - Strong synthetic biology capabilities - Backed by major investors - Beauty-focused biotech positioning	The Cosmos Edge ✓ Multi-Molecule Platform (Ectoine + Gadusol + MAAs) ✓ Cost Leader (India-based Biotech Ops) ✓ Process IP Moat (Pathway + Purification Optimization) ✓ Day-One Industrial Scalability ✓ Strategic Align: Reef-safe, Clean Beauty, UV-first
Strategic Limitations - Single-Molecule Focus (Restricted to Ectoine) - High Cost Structure (Western Operations) - General Dermocosmetics Positioning	Strategic Limitations - Scale Constraints (Early-stage platform) - High OpEx (US-centric geography) - Limited Presence in UV Ingredient Layer	

While others commercialize ingredients, we are defining a new UV category via Platform Advantage (Multi-molecule), Process IP (Pathway optimization), and Day-One Scalability.

Revenue First. Regulatory Dominance Next.

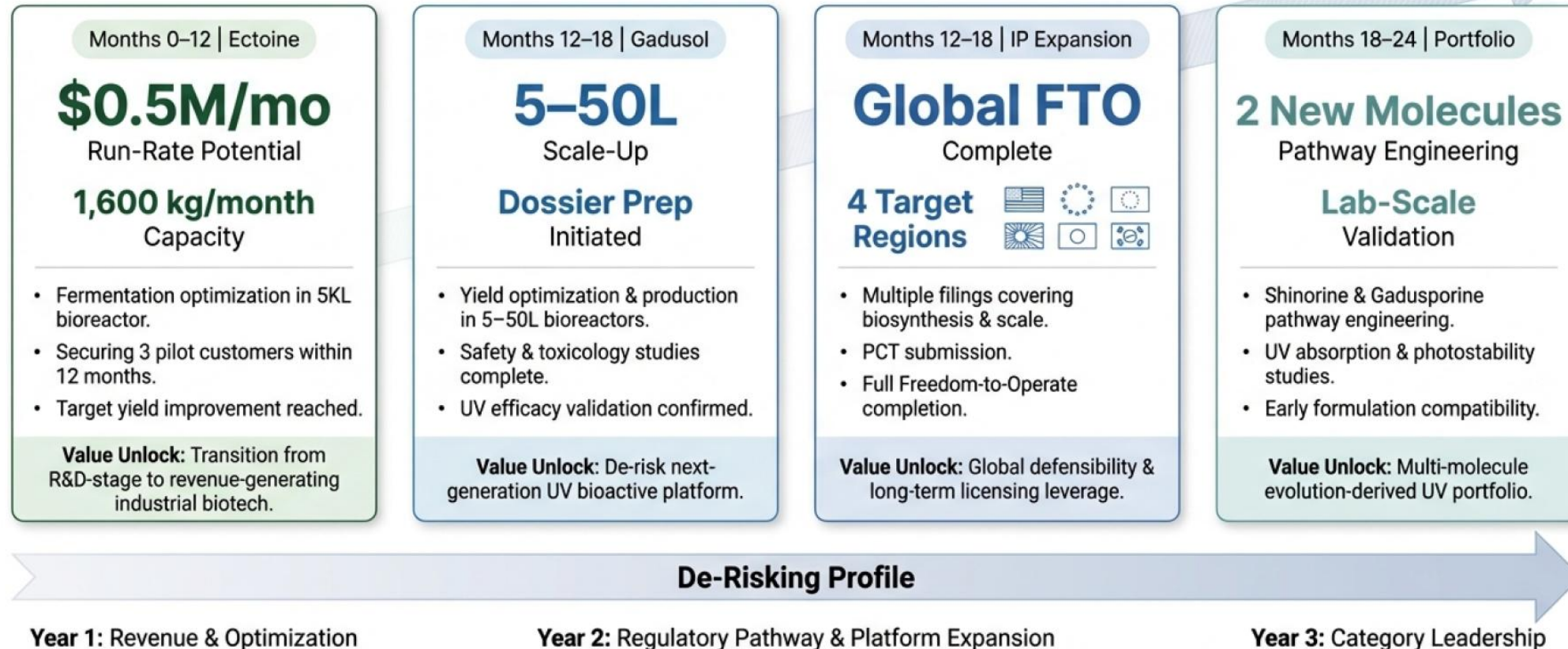
A structured, jurisdiction-dependent strategy to **minimize time-to-market risk** while securing long-term category leadership



18-24 Month Value Creation Roadmap

Strategic Outlook

From Industrial Optimization to Category Leadership



Execution today builds a scalable, IP-protected UV bioactive platform tomorrow.

The Ask – \$1.2Mn to Unlock Commercial Scale and Regulatory Positioning

Capital allocation focused on converting validated science into revenue-generating industrial capability and defensible IP.

Industrial Scale-Up (Ectoine)

\$0.5M



- 5KL fermentation optimization
- Yield enhancement & cost reduction
- Target capacity: 1,600 kg/month
- Pilot customer conversion & commercial readiness

Outcome: Revenue-generating industrial biotech capability.

Regulatory & Scale (Gadusol)

\$0.4M



- Scale-up from 5L to 50L bioreactors
- Safety & toxicology studies
- UV efficacy validation
- Regulatory dossier preparation

Outcome: De-risked next-generation UV bioactive pathway.

R&D Pipeline

\$0.1M

- Pathway engineering
- Lab-scale validation
- UV absorption & photostability testing

Outcome: Multi-molecule platform expansion.

IP & Compliance

\$0.1M

- PCT filing
- Global strategy (US, EU, JP, KR)
- Freedom-to-Operate analysis

Outcome: Global defensibility & regulatory positioning.

Market Dev

\$0.1M

- Commercialization support
- Customer acquisition & sampling
- Partnership development

Outcome: Revenue acceleration & brand positioning.

Investment Thesis: This \$1.2M converts core science into a commercial engine. Each dollar directly unlocks a milestone. Each milestone compounds valuation.

Team



Bharat Somireddy,
Ph.D.
CEO & Founder



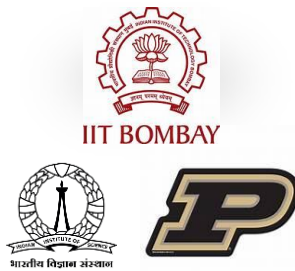
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